

# Doctor Directory Web Scraping: A Data Science Project Report

## 1. Objective

This project involved building an automated Python scraper (Requests + BeautifulSoup) to collect structured doctor-directory data from doctor.bd. The scraper paginated through listing pages, visited each doctor's profile page for additional fields, removed duplicates, and exported a UTF-8 CSV dataset.

## 2. Dataset Overview

The final dataset, `doctors_data.csv`, contains 2,000 unique doctor records across exactly eight columns: `doctor_name`, `specialty`, `qualification`, `designation`, `hospital`, `location`, `experience`, and `profile_url`. Each record was verified to correspond to a unique `profile_url`, confirming that the duplicate-removal step during scraping was fully effective — no duplicate URLs or duplicate full rows remain in the final file.

| Metric                              | Value        |
|-------------------------------------|--------------|
| Total records                       | 2,000        |
| Total columns                       | 8            |
| Unique profile URLs                 | 2,000 (100%) |
| Unique specialties represented      | 43           |
| Unique locations (districts/cities) | 35           |
| Unique hospitals / workplaces       | 399          |

## 3. Data Completeness

Most fields are essentially complete: `doctor_name`, `specialty`, `qualification`, and `location` have no missing values, while `designation` and `hospital` are each missing for a single record. The `experience` field is the exception — only 108 of 2,000 records (5.4%) report a numeric years-of-experience value, since the source site itself omits this figure for most profiles.

## 4. Key Findings

Dhaka leads with 958 records (47.9%), followed by Chattogram (12.5%) and Sylhet (6.4%); the top five locations make up 77.0% of the dataset, showing a strong urban skew.

By specialty, Gynaecology & Obstetrics tops the list (13.7%), followed by Pediatrics, Medicine, Cardiology, and Orthopedics — together 42.6% of records, with the rest spread across 38 other specialties.

By designation, Consultant is most common (29.7%), followed by Assistant Professor (23.8%) and Associate Professor (13.5%). Chattogram Medical College Hospital, Dhaka Medical College Hospital, and Bangladesh Medical University are the top listed workplaces.

Also, 68 doctor names repeat, since doctor.bd creates a separate profile per specialty or workplace, not per person.

## 5. Top Specialties and Locations

| Specialty                | Count |
|--------------------------|-------|
| Gynaecology & Obstetrics | 273   |
| Pediatrics               | 163   |
| Medicine                 | 159   |
| Cardiology               | 142   |
| Orthopedics              | 116   |

| Location   | Count |
|------------|-------|
| Dhaka      | 958   |
| Chattogram | 250   |
| Sylhet     | 128   |
| Khulna     | 113   |
| Cumilla    | 90    |

## 6. Conclusion

The scraping pipeline successfully produced a clean, de-duplicated dataset of 2,000 doctor records with the required eight-column structure and UTF-8 encoding. The data confirms that doctor.bd's directory is geographically concentrated in Dhaka and a small number of major cities, and that its specialty mix is dominated by Gynaecology & Obstetrics, Pediatrics, Medicine, Cardiology, and Orthopedics. The main data-quality limitation is the experience field, which the source website itself leaves blank for the vast majority of listed doctors.

## 7. Data Source

**Source website:** <https://doctor.bd/doctors>

Dataset file: doctors\_data.csv (2,000 records, submitted alongside this report)